

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 10+ years of experience building production applications in collaborative engineering environments. Offers strong evidence in JavaScript, TypeScript, SQL, PostgreSQL, Azure, and Docker, with additional hands-on background in .NET, ASP.NET, C#, and Python. Experience spans APIs, data-intensive systems, automation platforms, and cloud-based application delivery.

Selected Skills

Azure SQL, OpenAI, Tailwind CSS, AI Agents, Azure, Azure App Service, Azure DevOps, Azure Functions, Azure Storage Queues, CI/CD, Containerization, CSS, Docker, Docker Compose, FastAPI, HTML, JavaScript, Node.js, PostgreSQL, Redis, REST APIs, SCSS, SQL, SQL Server

Professional Experience

Enerex - AI Development Consultant

2023-11 - present

Architected frontend and AI service layers for task-based automation systems and cross-product integrations.

- Provisioned and deployed Azure Functions, web applications, databases, and Storage Queues to support Enerex products.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users.
- Designed the add-in to open directly from contract emails, analyze attached contracts with an agent carrying deep business knowledge, and return standardized extracted data.
- Implemented deployment pipelines for application delivery and environment updates.
- Built a review UI that allowed users to inspect and modify extracted contract data before submission.

First Due - Senior Software Engineer

2017-07 - 2019-11

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Architected a system to manage and control web crawlers, centralizing execution control and operational monitoring in a single UI.
- Improved control and accountability of company web-crawling operations through centralized management and visibility.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Implemented and ran SQL queries to investigate, validate, and repair data issues.
- Migrated projects from Visual Basic to ASP.NET as part of legacy application modernization work.
- Implemented numerous stored procedures to support application behavior and database operations.
- Performed extensive database data consistency reviews and corrections across business-critical datasets.

Broadcasting Platforms - Senior Frontend Engineer & Team Lead

2019-11 - 2022-12

Led and built real-time streaming web applications for performers and end-users using modern frontend, backend jobs, and event-driven systems.

- Used Redis as a non-relational data store for the in-app chat system.
- Worked in Linux environments while building and operating streaming platform systems.
- Served as team leader for frontend and platform delivery work.
- Acted as scrum master, coordinating team execution and agile delivery processes.
- Developed and maintained production systems across PHP, JavaScript, and Vue using Docker Compose, Vue.js, Bootstrap, and ESLint.

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Implemented AI bots integrated with Microsoft Teams.
- Designed and implemented tens of Dataverse tables to support the application's data model.
- Created dozens of Power Automate flows and dataflows to support compliance workflows and data movement.
- Developed a React application embedded inside a model-driven Power Apps application.
- Built a complex User Access Review compliance system on the Power Platform for NBCUniversal.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.
- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Deployed the system in Azure with CI/CD through Azure DevOps.
- Implemented AI-powered invoice analysis using OpenAI.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | 2004 - 2009